Assignment 4 – Regression Analysis

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BAN 100 – Statistic for Analysis

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Table of Contents

Problem 1 (bikes_sharing.csv) – 3

a. Multiple Regression Model – 3
b. Interpret the values of the coefficients in the model – 4
c. Test the model to see if it is significant – 5
d. Plot the residuals vs actual values – 6
e. Interpret the value of R^2 – 8
f. Do you think this model will be useful in helping the planners? – 8
g. Test the individual regression coefficients – 8
h. Which variables are more likely to drop from the model – 9
i. Stepwise regression – 9
j. Analyze the model – 10
k. Memo letter to boss – 11

Problem 2 (Titanic.xlsx) – 12

a. Logistic Regression Equation Relating Age and Survived – 12
b. Estimate the probability of surviving the passenger with the average Age 30 – 12
c. Check who have a 0.50 or higher probability of surviving – 13
d. Estimated odds ratio - 14
e. Comprehensive logistic regression model - 14
f. Interpret the model – 16
g. Provide the syntax and output - 17

Problem 3 (Capital punishment) – 18

a. Why the odds ratios are different? – 18
b. Show the relation between the odd ratios and coefficient – 18
c. What are the differences between model 1 and model 2? Key conclusion? – 18

Works Cited - 19

Problem 1: (bikes_sharing.csv)

In this dataset, bikes_sharing.csv which talks about bike rental data from Washington DC's Capital Bikeshare system. How it works is by tracking trip-details like date/time, departure context and usage patterns. It includes for calendar-based variables like Season, Holiday, and Working day. It captures weather conditions by using four categorized levels. It provides normalized environmental measures It distinguishes between casual users and registered users as dependent variables. However, the main dependent variables that will be using as regression is count which means total bike rental count.

a. Multiple Regression Model:

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$

$\beta_0 = estimate\ of\ the\ regression\ constant$

$\beta_k = estimate\ of\ regression\ coefficient\ k$

$\beta_1 = estimate\ of\ regression\ coefficient\ 1$

$\beta_2 = estimate\ of\ regression\ coefficient\ 2$

Dependent Variable:

- casual
- registered
- count (This variable will be using as dependent for regression analysis)

Independent Variable:

- datetime
- season
- holiday
- weather
- temp
- atemp
- humidity
- windspeed

Actual Regression Equation:

$count = \beta_0 + \beta_1 datetime + \beta_2 season + \beta_3 holiday + \beta_4 weather + \beta_5 temp + \beta_6 atemp + \beta_7 humidity + \beta_8 windspeed + \varepsilon$

b. <u>Interpret the values of the coefficients in the model:</u>

Regression Model for Bikes Sharing

The REG Procedure
Model: MODEL1
Dependent Variable: count

Number of Observations Read	10886
Number of Observations Used	10886

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	110867606	12318623	543.95	<.0001
Error	10876	246305308	22647		
Corrected Total	10885	357172914			

Root MSE	150.48814	R-Square	0.3104
Dependent Mean	191.57413	Adj R-Sq	0.3098
Coeff Var	78.55348		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	-3986.45790	147.78841	-26.97	<.0001
datetime	1	0.21937	0.00785	27.94	<.0001
season	1	3.11638	1.54707	2.01	0.0440
holiday	1	-8.20893	8.95481	-0.92	0.3593
workingday	1	-0.79286	3.20252	-0.25	0.8045
weather	1	4.09872	2.53100	1.62	0.1054
temp	1	1.27596	1.10344	1.16	0.2476
atemp	1	5.86738	1.01487	5.78	<.0001
humidity	1	-2.87270	0.08971	-32.02	<.0001
windspeed	1	1.01837	0.19337	5.27	<.0001

Figure 1: Regression Model for Bikes Sharing in SAS Output

- The intercept of this regression model is -3986.45790
- Regression Coefficients for datetime is 0.21937. This means that change of count by 0.21937 for every one unit increase in datetime while other independent variable is fixed.
- Regression Coefficients for season is 3.11638. This means that change of count by 3.11638 for every one unit increase in season while other independent variable is fixed.
- Regression Coefficients for holiday is -8.20893. Therefore, the count of bike rental is considered inversely proportional with holiday due to negative coefficient. This means that change of count of 8.20893 decrease in holiday while other independent variable is fixed.

- Regression Coefficients for workingday is -0.79286. Therefore, the count of bike rental is considered inversely proportional with workingday due to negative coefficient. This means that change of count of 0.79286 decrease in holiday while other independent variable is fixed.
- Regression Coefficients for weather is 4.09872. This means that change of count by 4.09872 for every one unit increase in weather while other independent variable is fixed.
- Regression Coefficients for temp is 1.27596. This means that change of count by 1.27596 for every one unit increase in temp while other independent variable is fixed.
- Regression Coefficients for atemp is 5.86738. This means that change of count by 5.86738 for every one unit increase in atemp while other independent variable is fixed.
- Regression Coefficients for humidity is -2.87270. Therefore, the count of bike rental is considered inversely proportional with humidity due to negative coefficient. This means that change of count of 2.87270 decrease in humidity while other independent variable is fixed.
- Regression Coefficients for windspeed is 1.01837. This means that change of count by 1.01837 for every one unit increase in windspeed while other independent variable is fixed.

c. <u>Test the model to see if it is significant</u>

$H_0: \beta_i = 0$

$H_a: \beta_i \neq 0$

Null Hypothesis (H_0): The value in this regression model does not fit in the bike sharing data

Alternative Hypothesis (H_a): The value in this regression model does fit in the bike sharing data

$Level\ of\ significance\ (alpha): \alpha = 0.05$

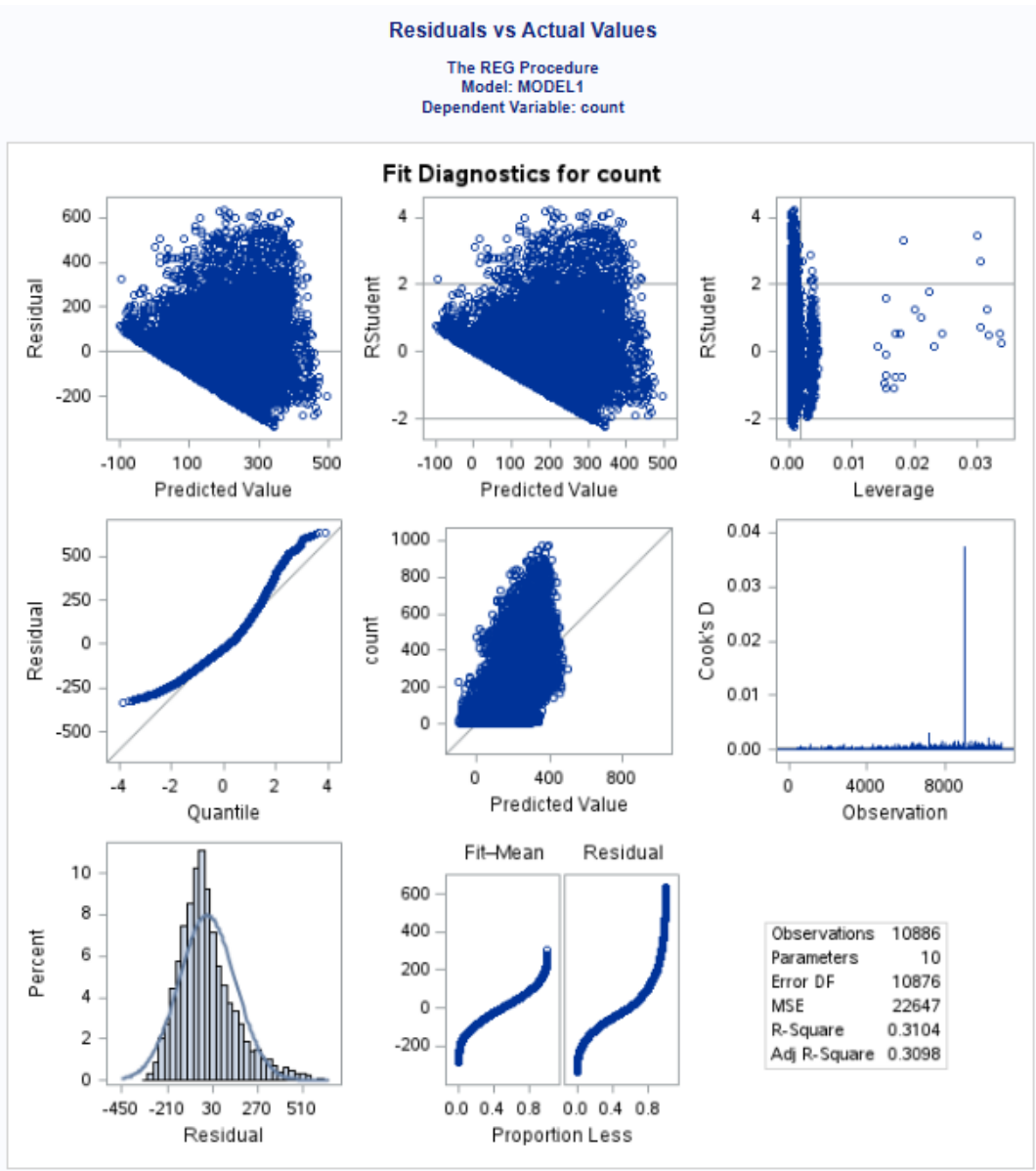
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	110867606	12318623	543.95	<.0001
Error	10876	246305308	22647		
Corrected Total	10885	357172914			

Figure 2: Analysis of Variance from Regression Model in SAS Output

According to *Figure 2: Analysis of Variance from Regression Model in SAS Output* for testing the model to see if it is significant where it focuses on "Pr > F" sections and the

result is saying "<0.001" and it is less than alpha of 0.05. Therefore, we reject the null hypothesis and conclude that variables are significant with the count of bike rentals.

d. <u>Plot the residuals vs actual values</u>



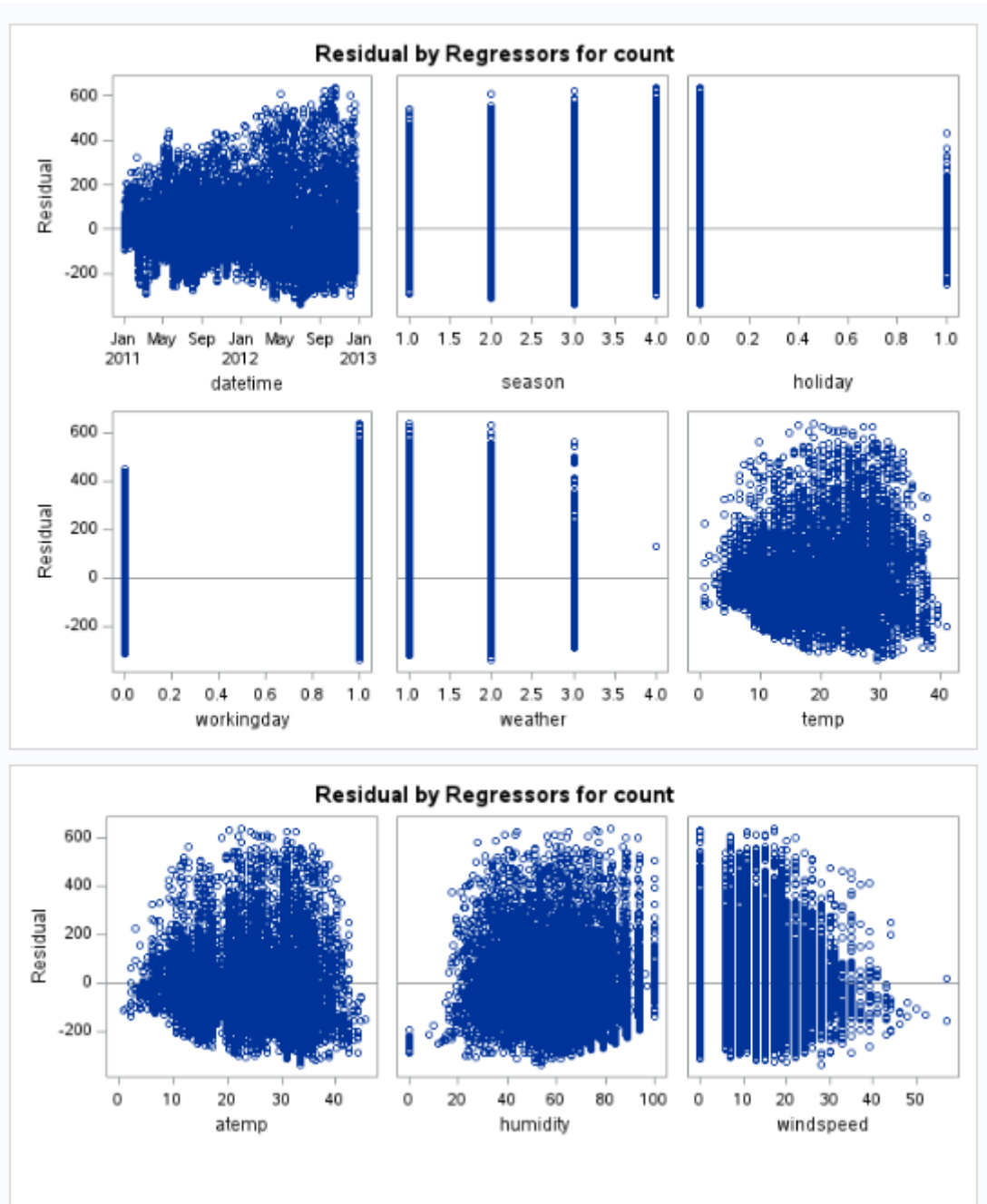


Figure 3: SAS Output of Plotting the Residuals vs Actual Values

Based on the plot, it suggests that it is considered as nonconstant error variance because it is showing that errors do not have a constant variance. Therefore, there are few of predictor could indicate an inadequate model while the predictor of season, holiday, workingday, and weather shows vertical straight lines which could possibly have undue influence. Therefore, the model does not do a good job of predicting number of bikes.

e. Interpret the value of R^2

Root MSE	150.48814	R-Square	0.3104
Dependent Mean	191.57413	Adj R-Sq	0.3098
Coeff Var	78.55348		

Figure 4: Second Part of Analysis of Variance in Regression Model SAS Output

From *Figure 4: Second Part of Analysis of Variance in Regression Model SAS Output* where the value of R-Square is 0.3104 which is 31.04%. Therefore, R-Square results of 0.3104 or 31.04% shows that the model is not a good fit in predicting the number of bikes.

f. Do you think this model will be useful in helping the planners?
Therefore, I think this model will not be useful in helping the planners because of the plots of residual variables as mentioned in part d and R-Squared in part e which shows that the model is not a good fit in predicting the count for dependent variable for number of bikes. However, individual independent variables might be useful in helping the planners in predicting the number of bikes.

g. Test the individual regression coefficients

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	-3986.45790	147.78841	-26.97	<.0001
datetime	1	0.21937	0.00785	27.94	<.0001
season	1	3.11638	1.54707	2.01	0.0440
holiday	1	-8.20893	8.95481	-0.92	0.3593
workingday	1	-0.79286	3.20252	-0.25	0.8045
weather	1	4.09872	2.53100	1.62	0.1054
temp	1	1.27596	1.10344	1.16	0.2476
atemp	1	5.86738	1.01487	5.78	<.0001
humidity	1	-2.87270	0.08971	-32.02	<.0001
windspeed	1	1.01837	0.19337	5.27	<.0001

Figure 5: Parameter Estimates from Regression Model in SAS Output

Alpha = 0.05 level of significance

H_0: $Variable\ coefficient\ are\ not\ significant\ predicting\ the\ number\ of\ counts$
H_a: $Variable\ coefficent\ are\ significant\ predicting\ the\ number\ of\ counts$

p-values of individual variables based on (Pr > |t|) sections:

If the p-values of individual variables < 0.05:
- datetime
- season
- atemp
- humidity
- windspeed

Therefore, we reject the null hypothesis and conclude that these variables are significant in predicting the number of bike rentals.

If the p-values of individual variables > 0.05:
- holiday
- workingday
- weather
- temp

Therefore, we failed to reject the null hypothesis and conclude that these variables are not significant in predicting the number of bike rentals.

h. Which variables are more likely to drop from the model?

I would choose holiday, workingday, weather, and temp because as mentioned in part g as these variables are not significant in predicting the number of bike rentals as if dropping these four variables who are not significant, it might have better model fit.

i. Stepwise regression

All variables left in the model are significant at the 0.1500 level.

No other variable met the 0.1500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	temp		1	0.1556	0.1556	2435.58	2005.53	<.0001
2	humidity		2	0.0855	0.2411	1089.43	1225.78	<.0001
3	datetime		3	0.0655	0.3066	57.9518	1028.38	<.0001
4	atemp		4	0.0015	0.3081	36.8572	23.03	<.0001
5	windspeed		5	0.0019	0.3100	9.0389	29.81	<.0001
6		temp	4	0.0001	0.3099	8.4747	1.44	0.2309
7	season		5	0.0002	0.3101	6.8163	3.66	0.0558
8	weather		6	0.0002	0.3103	6.0989	2.72	0.0993

Figure 6: SAS Output of Stepwise Regression

Based on the stepwise regression, in order to find the best model for the data, it would be better to drop variable temp due to p-value of 0.2309 which is greater than significance level at 0.15.

j. Analyze the model

As mentioned in part i based on the stepwise regression, in order to have the best fit model is to drop variable temp due to p-value is higher than the alpha of 0.05 and 0.15. Based on the plot, it shows that it is considered as nonconstant error variance due to error that does not have a constant variance. However, four of the predictor showing vertical straight lines which can be undue influence. The problem with this model that removal of variables that are not that significant to be removed by checking the dataset's R-squared value.

k. <u>Memo letter to boss</u>
Date: December 12, 2025
To: Manager
From Kevin Lu
Subject: Bike Sharing Dataset

In regarding to bike sharing dataset for predicting the counts as the number of bikes based on other variables. The findings that noted for the best model to use:

- Plotting the residuals vs values of the variables showing not a good fit because points not showing fairly across all the values of independent variables/
- R-Squared shows that it is not a good model fit which is showing 0.3104 or 31.04% showing below the industry of norm.
- Did testing with individual regression coefficients which helps to identify which one prove to be significant or insignificant. Five of the variables are datetime, season, atemp, humidity, and windspeed all prove to be significant because all the p-values of individual variables is less than alpha = 0.05. While, four of variables are holiday, workingday, weather, and temp showing insignificant in predicting the number of bike rentals because all the p-values are greater than alpha = 0.05.
- It is recommended to drop variables that are not significant in predicting the number of bike rentals however, the R-Squared may not increase if trying to drop insignificant variables.
- Used a stepwise regression suggest dropping variable temp in order to have best fit model due to higher p-value and it is greater than the set significance level. However, removing this variable may not increase the R-Square.

Therefore, using multiple regression would be great tool to analyze relationships based on variables, however, the assumption that there is a straight-line relationship between independent and dependent variables is not applicable to this dataset. For the best model but using individual regression coefficient can help to identify variables that are related or have significant to prove the effect in predicting the number of bikes.

Thank you!

Kevin Lu

Problem 2: (Titanic.xlsx)

In this dataset, there are 891 passengers on board. The goal for this data analysis is to predict the survival (survival = 1 indicates that it is survived passengers) based on passenger characteristics. Survival is one of dependent variables and it indicates as binary outcome variable (0 = did not survived, 1 = survived). The independent variable is Age, Sex (it converted from category to numerical variable) where male = 0, female = 1, Class, SiblingSpouse, ParentChild, and Embarked (it converted into dummy variables) where S = 1, C = 2, Q = 3. Therefore, the objective to analyze this dataset is to build logistic regression model to estimate the odds of passenger's probability of survival.

a. Logistic Regression Equation Relating Age and Survived

$$f(x) = p = \frac{e^{\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k}}{1 + e^{\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k}}$$

$$f(x) = p = \frac{e^{\beta_0 + \beta_1 Age}}{1 + e^{\beta_0 + \beta_1 Age}}$$

$$logit(p_i) = \beta_0 + \beta_1 Age$$

$$\beta_0 = Slope\ Intercept$$
$$\beta_1 = Age$$

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-0.0567	0.1736	0.1068	0.7438
Age	1	-0.0110	0.00533	4.2310	0.0397

Figure 7: SAS Output of Analysis of Maximum Likelihood Estimates for Logistic Regression

b. Estimate the probability of surviving the passenger with the average Age 30

$$f(x) = p = \frac{e^{-0.0567 + (-0.0110)Age}}{1 + e^{-0.0567 + (-0.0110)Age}}$$

$$f(x) = p = \frac{e^{-0.0567 + (-0.0110)(30)}}{1 + e^{-0.0567 + (-0.0110)(30)}}$$

$$f(x) = p = \frac{e^{-0.3867}}{1 + e^{-0.3867}}$$

$$f(x) = p = \frac{0.679294852}{1.679294852}$$

$$f(x) = p = 0.405 \;\; or \; 40.5\%$$

Therefore, the estimated probability of surviving the passenger with the average Age 30 is roughly 40.4%.

Obs	Age	P_1
1	30	0.40477
2	30	0.40477

Figure 8: SAS Output of estimated probability of surviving the passenger with the average Age of 30.

c. <u>Check who have a 0.50 or higher probability of surviving</u>

$$0.50 = \frac{e^{-0.0567 + (-0.0110)Age}}{1 + e^{-0.0567 + (-0.0110)Age}}$$

$$Age = \frac{In\left(\frac{0.50}{1 - 0.50}\right) - \beta_0}{\beta_1}$$

$$Age = \frac{-\beta_0}{\beta_1}$$

$$Age = \frac{0.0567}{-0.0110}$$

Age = -5.1545

Average Age to achieve the level of probability of 0.50

Obs	p	Intercept	Age	Average_age
1	0.5	-0.0567	-0.011	-5.15455

Figure 9: SAS Output of Average Age to achieve the level of probability of 0.50

d. Estimate odds ratio:

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Age	0.989	0.979	0.999

Figure 10: SAS Output of Estimate Odds Ratio

$$Point\ Estimate = e^{-0.011} = 0.98906027 = 0.989$$

$$Odds\ of\ survival = 1 - 0.989$$

Odds of survival = 0.01093973

Odds of survival = 1%

Therefore, every one-year increase in age, the odds of survival decrease by estimate of 1%, holding other factors constant.

e. Comprehensive logistic regression model

Variables to do comprehensive logistic regression model:

Dependent:

- Survived

Independent:

- Age
- Class
- Sex
- SiblingSpouse
- ParentChild
- Fare
- Embarked

$$p = \frac{e^{\beta_0 + \beta_1 Age + \beta_2 Class + \beta_3 Sex + \beta_4 SiblingSpouse + \beta_5 ParentChild + \beta_6 Fare + \beta_7 Embarked}}{1 + e^{\beta_0 + \beta_1 Age + \beta_2 Class + \beta_3 Sex + \beta_4 SiblingSpouse + \beta_5 ParentChild + \beta_6 Fare + \beta_7 Embarked}}$$

$$logit(p) = \beta_0 + \beta_1 Age + \beta_2 Class + \beta_3 Sex + \beta_4 SiblingSpouse + \beta_5 ParentChild + \beta_6 Fare + \beta_7 Embarked$$

The LOGISTIC Procedure

Model Information	
Data Set	WORK.TITANIC_1
Response Variable	Survived
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	891
Number of Observations Used	712

Response Profile		
Ordered Value	Survived	Total Frequency
1	0	424
2	1	288

Probability modeled is Survived='1'.

Note: 179 observations were deleted due to missing values for the response or explanatory variables.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	962.904	651.054
SC	967.472	687.599
-2 Log L	960.904	635.054

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	325.8493	7	<.0001
Score	284.0565	7	<.0001
Wald	191.8822	7	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	2.6328	0.6011	19.1843	<.0001
Age	1	-0.0440	0.00820	28.7774	<.0001
Class	1	-1.2392	0.1630	57.7639	<.0001
sex_numerical	1	2.6169	0.2204	141.0136	<.0001
SiblingSpouse	1	-0.3697	0.1272	8.4535	0.0036
ParentChild	1	-0.0580	0.1230	0.2225	0.6371
Fare	1	0.00193	0.00251	0.5885	0.4430
embarked_numerical	1	0.0975	0.1972	0.2446	0.6209

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Age	0.957	0.942	0.972
Class	0.290	0.210	0.399
sex_numerical	13.693	8.890	21.090
SiblingSpouse	0.691	0.539	0.886
ParentChild	0.944	0.741	1.201
Fare	1.002	0.997	1.007
embarked_numerical	1.102	0.749	1.622

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	85.8	Somers' D	0.717
Percent Discordant	14.1	Gamma	0.717
Percent Tied	0.0	Tau-a	0.346
Pairs	122112	c	0.859

Figure 11: SAS Output for Comprehensive Logistic Regression

f. <u>Interpret the model</u>

Based on the model as mentioned in Part g and SAS output from *Figure 11: SAS Output for Comprehensive Logistic Regression*:

- The intercept of this logistic regression model is 2.6328
- Coefficient of Age:
  - Coefficient of age is -0.0440
  - Odd Ratio Estimates of Age is 0.957
  - Significant due to <0.001 which is less than alpha = 0.05
  - Every year, Age reduces the odd of survived by 4.40%, therefore lower age (younger passengers are likely to survived)
- Coefficient of Class:
  - Coefficient of Class is -1.2392
  - Odd Ratio Estimates of Class is 0.290
  - Significant due to <0.001 which is less than alpha = 0.05
  - Each year, Class are much more likely to have lower odds of survival, therefore higher class are more likely to get survived.
- Coefficient of Sex:
  - Coefficient of Sex is 2.6169
  - Odd Ratio Estimate of Sex is 13.693
  - Significant due to <0.001 which is less than alpha = 0.05
  - Every year, females are more likely to have higher odds of survived than males, therefore, gender is one of the strong predictors
- Coefficient of SiblingSpouse:
  - Coefficient of SiblingSpouse is -0.3697
  - Odd Ratio Estimate of SiblingSpouse is 0.691
  - Significant due to 0.0036 which is still less than alpha = 0.05
  - More siblings or spouse could reduce the odd of survival therefore it might be more complications traveling because of large family groups
- Coefficient of ParentChild:
  - Coefficient of ParentChild is -0.0580
  - Odd Ratio Estimate of ParentChild is 0.944
  - The odds or survival decrease by 0.0580
  - Not significant due to 0.6371 greater than alpha = 0.05
- Coefficient of Fare:
  - Coefficient of Fare is 0.00193
  - Odd Ratio Estimate of Fare is 1.002
  - The odds of survival increase by 0.0580
  - Not significant due to 0.4430 greater than alpha = 0.05
- Coefficient of Embarked:

- Coefficient of Embarked is 0.0975
- Odd Ratio Estimate of Embarked is 1.102
- The odds of survival increase by 0.0975
- Not significant due to 0.6209 greater than alpha = 0.05
- Therefore, embarking does not mean affecting the survival.

Therefore, it is showing that this logistic regression has some strong model fit due to adding some variables other than age since there are some of variables that would affect the odds of survival.

g. Providing Syntax and Output

Part a: **Figure 7: SAS Output of Analysis of Maximum Likelihood Estimates for Logistic Regression**

Part b: **Figure 8: SAS Output of estimated probability of surviving the passenger with the average Age of 30**

Part c: **Figure 9: SAS Output of Average Age to achieve the level of probability of 0.50**

Part d: **Figure 10: SAS Output of Estimate Odds Ratio**

Part e: **Figure 11: SAS Output for Comprehensive Logistic Regression**

Problem 3: (Capital punishment)

Since this dataset had been already analyzed, it already runs two logistic regression model predicting capital punishment using race. However, the only difference that the race was coded like White = 0 and Black = 0. The odds ratio for White in Model 2 is roughly 2.948 while the odds ratio for Black in Model 1 is roughly 0.3392, however the coefficient, odds for white defendants and odds for black defendants are the same. However, the odd ratios are different as white defendants have higher odds of getting capital punishment than black defendants.

a. Why odds ratios are different?
The odds ratios are different because of ratios considered. As it showing in Model 1 odds ratio is 0.34 while the Model 2 odds ratio is 2.95 which implies that Model 1 odds ratio is less than 1 while Model 2 is greater than 1, shows lesser chance of happening for capital punishment in Model 1 while Model 2 shows greater chance of capital punishment.

b. Show relation between the odd ratios and coefficient

$$Odds\ Ratio = e^{\beta}$$

Model 1	Model 2
Odds ratio = $e^{-1.081}$	Odds ratio = $e^{1.081}$
Odds ratio = 0.339256099	Odds ratio = 2.947625703

Therefore, the odd ratio is just an exponential of coefficient.

c. What are the differences between model 1 and model 2? Key conclusion?
Differences:
- Coefficient changes (-1.081 vs 1.081)
- Odds ratios are reciprocals (0.34 vs 2.95)
- Reference categories are different in each model

Key conclusion:

Therefore, the numbers look different, however, both models have the same story. Compared to white defendants, black defendants have lower odds of receiving capital punishment in Model 1. And compared to black defendants, white defendants have higher odds than black defendants in Model 2.

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